

06	D
----	---

11	B
----	---

12	c
----	---

13	A
----	---

14	B
----	---

15	c
----	---

16	D
----	---

17	B
----	---

18	D
----	---

13	B
----	---

15	D
----	---

16	A
----	---

17	B
----	---

18	D
----	---

19	B
----	---

20	B
----	---

Question	Answers	Additional comments/Guidelines	Mark	AO
01.1	Two from: ✓ ✓ negligible forces between particles (except during collisions) volume of particles is negligible (compared to volume of container) particles collide elastically collision duration is negligible compared to time between collisions	Accept “point particles/molecules” Accept particles move randomly	2	AO1 AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
01.2	Converts T to kelvin (285.5 K) ✓ Uses $\frac{1}{2}m(c_{\text{rms}})^2 = \frac{3}{2}kT$ ✓ 423 (m s^{-1}) ✓ Any 2 sf answer from some appropriate working ✓	Allow other formulae involving $(c_{\text{rms}})^2$ Using 12.5 °C for T gives 88.5 (m s^{-1})	4	AO1 AO2 AO2 AO4
Total			6	

Question	Answers	Additional comments/Guidelines	Mark	AO
----------	---------	--------------------------------	------	----

Question	Answers	Additional comments/Guidelines	Mark	AO
01.1	Use of $pV = NkT$ ✓ ($N =$) 3.73×10^{23} to at least 3 sf ✓	Alternative 1 st mark $pV = nRT$ and $N = nN_A$ Allow one mark for $n = 0.62$ if no other mark awarded	2	2 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
01.2	Use of (total) mass = $N \times$ mass of one particle AND density = mass / volume ✓ density = $0.18 \text{ (kg m}^{-3}\text{)}$ ✓	allow POT error for MP1 $0.17 \text{ (kg m}^{-3}\text{)}$ if Show value used	2	2 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
01.3	Statement of 1 st law with at least two terms defined (in the context of the gas in the container) ✓ Gas does work while expanding (against atmospheric pressure and the balloon's elastic) ($W < 0$) ✓ Little or no heat transfer into the balloon ($Q \sim 0$) ✓ Internal energy decreases and links this directly to temperature ($\Delta U < 0$) ✓	expect statement of 1 st law in symbols or words, given with context accept for MP3: less heat is transferred into the gas than work done MP2-4 must refer to gas in the balloon for MP4, link must refer to KE or an equation	4	1 × AO1 3 × AO2

Total			0	
-------	--	--	---	--

Question	Answers	Additional comments/Guidelines	Mark	AO
----------	---------	--------------------------------	------	----

Question	Answers	Additional comments/Guidelines	Mark	AO
06.1	Compares state 1 and state 2 using an ideal gas equation ✓ Provides evidence that the condition $d_2 > d_1$ or $V_2 < V_1$ leads to $\frac{T_2}{T_1} < 40$ ✓	Expect to see, after some working: $\frac{d_2 T_2}{d_1 T_1} = \frac{p_2}{p_1} = 40 \quad \text{OR} \quad \frac{V_1 T_2}{V_2 T_1} = \frac{p_2}{p_1} = 40$ For full credit there must be a link between density and volume in the answer.	2	AO2 AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
06.2	Work is done on air (by compressor blades) ✓ Little time for heat transfer so $Q \sim 0$ ✓ Uses first law to explain why internal energy increases or why ΔU is positive (and so temperature increases) ✓	Mention of “adiabatic” process scores MP2.	3	AO2 AO2 AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
06.3	Substitution into formula ✓ Manipulation of formula (either using logs or roots) ✓ 860 (K) ✓	MP2: Condone manipulation of an incorrect substitution e.g. $40 = \left(\frac{300}{T_2}\right)^{3.5}$	3	AO3 AO4 AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
06.4	Use of $Q = \frac{kA\Delta\theta}{l}$ for either material ✓ Correct substitution e.g. $\frac{0.74(1600 - \theta_1)}{250 \times 10^{-6}} = \frac{27(\theta_1 - 970)}{2.0 \times 10^{-3}}$ ✓ 1100 (K) ✓	Condone mismatch between k and $\Delta\theta$ in MP1 but not in MP2. Condone power of ten errors for l in MP1 and MP2. 3 sf answer is 1080 (K)	3	AO2 AO2 AO2

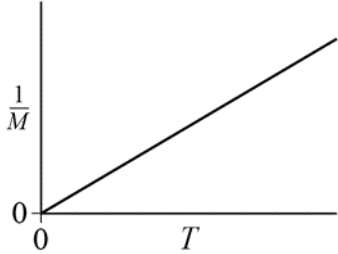
Question	Answers	Additional comments/Guidelines	Mark	AO
06.5	Converts rpm to rad s^{-1} ✓ Uses $P = T\omega$ ✓ $1.8 \times 10^4 \text{ (N m)}$ ✓	MP1: expect 1050 rad s^{-1} Answer of 1.93×10^3 gains MP2. Allow 2 marks for using revs per second, giving $1.14 \times 10^5 \text{ (N m)}$	3	AO2 AO2 AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
06.6	Attempt to get area under curve ✓ Counts squares (39–41 'big' squares) OR determines energy per square ('big' = 2.5 MJ, 'little' = 0.1 MJ) OR uses a geometrical method ✓ Answer between 0.97×10^8 and $1.03 \times 10^8 \text{ (J)}$ ✓	MP1: Area must be between 35 m^3 and 75 m^3 Accept 1 sf answer of $1 \times 10^8 \text{ (J)}$	3	AO3 AO3 AO3
Total			17	

Question	Key	Answer	AO
----------	-----	--------	----

7	A	-45 -60	AO1
---	---	---------	-----

9	D	the ideal gas equation	AO1
---	---	------------------------	-----

10	B		AO1
----	----------	---	-----

11	D	3.1×10^{28}	AO1
----	---	----------------------	-----

12	c	$\frac{v}{\sqrt{2}}$	AO1
----	----------	----------------------	-----

14	A	increase increase	AO1
----	---	------------------------	-----

Question	Answers	Additional comments/Guidelines	Mark	AO
01	Max 4 from:	Mark as a list. Mark the answer space as a whole: ignore the '1' and '2' Max 2 from each column. The 'property' is about the gas molecules, the 'observation' is about the smoke particles. Accept alternatives for 'random' such as 'unpredictable', 'no preferred direction', 'various directions', 'irregular' etc. Allow 'jiggle' for 'move randomly'. An unqualified reference to 'particles' is taken to mean the smoke particles. An unqualified 'they' refers to the gas molecules.	4	2 × AO1 2 × AO3
	Property✓✓			
	move in random directions/randomly			
	very small/negligible volume			
	move very fast			
	Random speeds/ range of KE			
Total			4	

Question	Answers	Additional comments/Guidelines	Mark	AO
----------	---------	--------------------------------	------	----

Question	Answers	Additional comments/Guidelines	Mark	AO
02.1	Two from ✓✓ Converts temperature to kelvin Evidence of attempt to use $pV = nRT$ Evidence of attempt to use ($V=$) $\pi r^2 h$ $r = 0.12$ (m) ✓	accept 0.13 (m)	3	1 × AO1 1 × AO2 1 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
02.2	(Read off and subtraction =) 95 – 25 OR 70 (mol) seen OR Evidence of use of mass = their amount (mol) × 0.029 (kg mol ⁻¹) OR mass = their amount (mol) × 29 (g mol ⁻¹) ✓ mass = 2.0(3) kg ✓	Accept 71 to 69 Expect 70 × 0.029 = 2.0(3) kg Reject 1 sf answer Accept answers between 2.0 and 2.1 (kg)	2	1 × AO3 1 × AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
02.3	Calculates $\frac{\Delta p}{\Delta n}$ or $\frac{p}{n}$ for at least two points and shows that it is constant. _{1✓} Uses data from Figure 3 to show that the intercept is zero. _{2✓} Refers to $pV = nRT$, with R and V constant (and therefore T is constant). _{3✓}	MP1 and MP2 can be awarded for calculating $\frac{p}{n}$ (expect 4×10^4 Pa/mol) OR T (expect 290 (K)) and showing it is constant for at least two points. In MP3 accept 'V/R constant' for 'R and V constant' Alternative: Relates $pV = nRT$ to $y = mx + c$ to show that gradient is $\frac{RT}{V}$ _{1✓} Uses data from Figure 3 to show that $c = 0$ _{2✓} R and V constant (therefore T constant) _{3✓}	3	1 × AO1 1 × AO2 1 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
02.4	Internal energy = sum of kinetic energies (and potential energies) of molecules ✓ PE zero (as gas ideal) ✓ Average KE constant as T is constant ✓ Therefore internal energy increases as n increases ✓	Accept (internal) energy = $\frac{3}{2}nRT$ OR $\frac{3}{2}NkT$. Condone missing reference to molecules No reference to potential energy gets Max 3 In MP3 condone “average speed” for “average KE”. Condone ‘total’ for ‘internal’	4	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
02.5	Steeper <u>straight</u> line ✓ Higher starting pressure ✓		2	AO2
Total			14	

Question	Answers	Additional comments/Guidelines	Mark	AO
----------	---------	--------------------------------	------	----

10	A	the number of molecules in the gas $\frac{R}{N_A}$	AO1
----	---	--	-----

11	B	They have negligible mass.	AO1
----	---	----------------------------	-----

Question	Answers	Additional comments/Guidelines	Mark	AO
03.1	Max 2 ✓✓ - Idea that $\Delta E_k = \Delta E_p$ $\Delta E_k = 1.6 \times 10^{-14} - 2 \times 4.0 \times 10^{-15} = 8 \times 10^{-15} \text{ (J)}$ - Correct substitution into $E_p = \frac{1}{4\pi\epsilon_0} \frac{Q^2}{r}$ $2.9 \times 10^{-14} \text{ (m)} \checkmark$	Expect to see $8.0 \times 10^{-15} = \frac{(1.6 \times 10^{-19})^2}{4\pi \times 8.85 \times 10^{-12} r}$ Condone missing substitution for ϵ_0	3	1 × AO1 1 × AO2 1 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
03.2	Max 2 ✓✓ - They can be moving in opposite directions / range of directions / a specified direction - Correct qualitative or quantitative statement about the (total) change of kinetic energy OR - Idea that closest approach is when kinetic energy is zero (if travelling in opposite directions). - Idea or use of distance of closest approach is inversely proportional to the (change in) kinetic energy Correct statement about the distance of closest approach for their directions / range of directions. (Must follow from some correct physics, although this may not be enough to gain the other marks.) ✓	If no direction (or range of directions) is given, first bullet point cannot be awarded but assume they are travelling in opposite directions for other bullet and marking points ΔE_k increases by a factor of 4 so d is quartered gains bullet 4 and MP3. Condone doubling $(\Delta)E_k$ to halve r for bullet 4 and MP3 Condone an incorrect number for MP3 if larger or smaller is correct. Ignore references to momentum. Expect something like Opposite directions ΔE_k is greater / 3.2×10^{-14} J Distance of closest approach is smaller Multiple directions are possible ΔE_k can have a range of values / 3.2×10^{-14} J to 0 Distance of closest approach may be larger or smaller (or the same) Close to parallel ΔE_k is (very) small / close to 0 Distance of closest approach is large(r)	3	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
03.3	Idea that there is a range of kinetic energies / velocities of particles (of a gas or plasma) so some nuclei will get close enough to fuse ✓	Accept the idea that other nearby nuclei may affect the distance of closest approach	1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
03.4	Any two from ✓✓ Kinetic theory requires no intermolecular forces / potential energy but this is not true in a plasma (because there are large electrostatic or magnetic forces / is potential energy between ions) Kinetic theory requires that the volume of the particles is insignificant compared with the volume of the gas but this is not true in a plasma (because the ions may be very close together) Kinetic theory requires that the duration of collisions is insignificant compared with the time between collisions but this is not true in a plasma (because the ions interact over a large range of distances)	Allow collisions are not elastic due to fusion occurring. Condone motion is not random due to (external) electric and/or magnetic fields (because gas particles are neutral and plasma particles are charged) Ignore any reference to non-identical particles.	2	AO4

Question	Answers	Additional comments/Guidelines	Mark	AO
03.5	Use of $E_k = \frac{3}{2}kT$ OR $\frac{3RT}{2N_A}$ ✓ 7.7×10^8 (K) ✓		2	AO1
Total			11	

Question	Answers	Additional comments/Guidelines	Mark	AO
----------	---------	--------------------------------	------	----

Question	Answers	Additional comments/Guidelines	Mark	AO
06.1	Attempts to use the area under the graph ✓ Counts 26 to 28 squares OR recognises that one (5x5) square is equivalent to 1.25 J OR Use of the trapezium rule with at least 4 trapeziums OR breaking into triangles and squares if at least 4 triangles along length of curve ✓ Answer in the range 32.5 (J) to 35 (J) ✓	Condone for mp2 (10 × 10) squares 6.5 to 7.0 squares OR one is equivalent to 5.0 J (1 × 1) squares 650 to 700 squares OR one is equivalent to 0.05 J	3	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
06.2	Uses $\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$ or correct use of $n = \frac{pV}{RT}$ or $N = \frac{pV}{kT}$ ✓ 440 OR 444 (K) ✓	Allow errors in read offs for MP1.	2	1 × AO1 1 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
06.3	Uses $pV = nRT$ to attempt to find the number of moles in the pump ✓ Additional quantity of air / their number of moles ✓ 26 or 27 if justified ✓	Expect to see $\frac{1 \times 10^5 \times 2.5 \times 10^{-4}}{8.3 \times 290}$ or $\frac{4.5 \times 10^5 \times 0.85 \times 10^4}{8.3 \times 440}$ or $\frac{4.5 \times 10^5 \times 0.85 \times 10^4}{8.3 \times \text{their } T \text{ in 6.2}}$ If $1.0(4) \times 10^{-2}$ (mol) seen award first 2 marks. Allow ecf only for incorrect T from 6.2	3	1 × AO1 1 × AO2 1 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
06.4	Work (W) is done on the gas ✓ No time or short time for any heat transfer (Q) ✓ Correct reference to the 1st law equation AND connects internal energy to temperature ✓	Ignore references to friction. Accept small value of ΔQ because time is short. Condone wrong direction of work in MP3 but not MP1.	3	1 × AO1 1 × AO2 1 × AO3
Total			11	

Question	Key	Answer	AO
----------	-----	--------	----

07	D	the potential energies and the translational, rotational and vibrational kinetic energies of all of the molecules.	AO1
			

11	A	1.0	2.8		AO2
----	---	-----	-----	--	-----

12	C	Air molecules are seen to move randomly.	AO1
----	---	--	-----

Question	Answers	Additional comments/Guidelines	Mark	AO
02.1	Uses $\frac{3}{2} kT$ with T in kelvin ✓ 3.5×10^{-22} (J) ✓	Accept ΔT in MP1	2	1 × AO1 1 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
02.2	Use of $p\Delta V$ ✓ 30(.25) (J) ✓		2	1 × AO1 1 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
02.3	Any two from: ✓✓ - converts temperature to K - uses $N = \frac{pV}{kT}$ OR $N = \frac{N_A pV}{kT}$ - subtracts their N_{up} from their N_{down} OR subtracts V_{up} from V_{down} 7.2×10^{21} ✓		3	1 × AO1 2 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
02.4	Idea that: external pressure is lower so moving air out of the balloons requires less energy ✓ more energy needed in heating the air as the air starts colder ✓ the total work done on the air as it leaves the balloon is the same if ΔN is the same (since $p\Delta V = \Delta NkT$) ✓		3	AO2
Total			10	

Question	Answers	Additional comments/Guidelines	Mark	AO
----------	---------	--------------------------------	------	----

7	A	0.20	AO3
---	---	------	-----

11	D	The volume of an individual particle is negligible.	AO1
----	---	---	-----

13	B	increase decrease	AO4
----	---	----------------------	-----

17	C	$7.8 \times 10^4 \text{ s}$	AO2
----	---	-----------------------------	-----

18	A	3.1×10^{-4}	AO3
----	---	----------------------	-----

19	B	4 cm^3	AO2
----	---	-----------------	-----

Question	Answers	Additional comments/Guidelines	Mark	AO
01.1	Any two from: ✓✓ - evidence of correct calculation of volume of container (= $1.79 \times 10^{-3} \text{ m}^3$) - evidence of calculation of volume of one sweet (= (their container volume – 1.15×10^{-3}) $\div$ 595) - calculates mass of one sweet (= their volume of one sweet $\times$ 1180) $1.3 \times 10^{-3} \text{ (kg)}$ ✓	Condone power of ten error in MP1 and MP2 $(6.39 \times 10^{-4} \div 595 =) 1.07 \times 10^{-6} \text{ m}^3$ Alternative BP2 and BP3 - evidence of calculation of mass of sweets = (their container volume – 1.15×10^{-3}) $\times$ 1180 (≈ 0.75) - calculates mass of one sweet (= their mass of sweets $\div$ 595) Do not condone missing subtraction in BP2.	3	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
01.2	Any one from: - calculates their number of moles using $pV = nRT$ - multiplies their number of moles by molar mass of nitrogen OR ✓ Any one from: - calculates their number of particles using $pV = NkT$ - divides their number of particles by Avogadro's number and multiplies by molar mass of nitrogen 1.5 (g) ✓	number of moles = 5.2×10^{-2} number of particles = 3.15×10^{22} Accept 3 sf or more answers that round to 1.46 or 1.47	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
01.3	Idea that $p \div T$ (in kelvin) is constant with their converted T OR Evidence of conversion to K e.g. 273 -15 OR 258 seen ✓ 9.8×10^4 (Pa) ✓	Allow use of $pV = nRT$ OR $pV = NkT$ using their n and V OR their N and V from 1.2 allow ecf for these routes in MP2 allow 9.7×10^4	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
01.4	Any two from: ✓✓ • (at lower temperature) gas particles inside container move more slowly • gas particles hit inside of container less often (per second) AND at lower mean speed/ lower mean change in momentum on collision • idea that the smaller (average change in) momentum/impulse means that a smaller force is exerted on lid/container (when cold than when warm) Force from gas outside > force from gas inside (so lid is harder to take off) ✓	Allow reverse argument based on warmer gas outside Allow missing reference to “mean” in BP2 if some relevant idea of average or mean seen elsewhere eg.reference to average KE If no other mark awarded allow one mark for argument based on the pressure being less when cold	3	AO2
Total			10	

Question	Answers	Additional comments/Guidelines	Mark	AO
----------	---------	--------------------------------	------	----